

The $u^{3/2}$ Endpoint Remainder for a Truncated Gaussian Heat Kernel

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Abstract

For $u > 0$, let S_u be the integral operator on $L^2([-1, 1])$ with kernel

$$\frac{1}{2\sqrt{\pi u}} \exp\left(-\frac{(p-q)^2}{4u}\right),$$

and let $\lambda(u)$ be its largest eigenvalue. We prove, with $A = \pi^2/4$, that there are constants $c, C, u_0 > 0$ such that

$$cu^{3/2} \leq \log \lambda(u) + Au \leq Cu^{3/2} \quad (0 < u < u_0).$$

Thus the signed first-order remainder has the pure-power scale $u^{3/2}$, the second Peano quotient tends to $+\infty$, and the finite classical right second derivative of $\log \lambda$ at zero does not exist. The lower bound comes from a cosine trial function with a uniform $O(u^2)$ Fourier-tail remainder; the upper bound comes from domination by a compressed resolvent whose principal eigenvalue is an explicit Robin eigenvalue. We do not claim an exact coefficient for the optimized $u^{3/2}$ remainder.

1 Statement and context

Let $H = L^2([-1, 1])$ and, for $u > 0$, define

$$(S_u f)(p) = \int_{-1}^1 \frac{e^{-(p-q)^2/(4u)}}{2\sqrt{\pi u}} f(q) dq. \quad (1)$$

The continuous symmetric kernel makes S_u compact and self-adjoint; its strict positivity gives a simple positive largest eigenvalue $\lambda(u)$. Set $\lambda(0) = 1$, $\Phi(0) = 0$, and $\Phi(u) = \log \lambda(u)$ for $u > 0$.

This operator arises as the continuum scaling operator in the finite-support quantum-walk problem of Liu and Chen [1]. They established

$$\lambda(u) = 1 - \frac{\pi^2}{4}u + o(u) \quad (u \downarrow 0) \quad (2)$$

and left higher-order endpoint analysis for future work. The sharp Fourier cutoff in (1) also places the problem in the broad tradition of time–frequency limiting operators [2]. At a more general methodological level, exactly solvable model classes are also used to certify bounds in constrained spectral-constant problems [3]; that work concerns spectral sets rather than heat kernels, and no result from it is used below.

Put

$$\kappa := \frac{\pi}{2}, \quad A := \kappa^2 = \frac{\pi^2}{4}, \quad R(u) := \Phi(u) + Au.$$

The following theorem resolves the endpoint question.

Theorem 1. *There are constants $c, C, u_0 > 0$, independent of u , such that*

$$cu^{3/2} \leq R(u) \leq Cu^{3/2} \quad (0 < u < u_0). \quad (3)$$

Consequently,

$$\frac{2[\Phi(u) + Au]}{u^2} \longrightarrow +\infty \quad (u \downarrow 0), \quad (4)$$

and the finite limit

$$\lim_{u \downarrow 0} \frac{\Phi'(u) + A}{u} \quad (5)$$

does not exist.

The theorem determines the correct signed pure-power scale. It does not assert the existence of a limit for $R(u)/u^{3/2}$.

2 A uniform trial-function lower bound

We use the Fourier transform

$$\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-ip\xi} f(p) dp, \quad \|f\|_2^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{f}(\xi)|^2 d\xi.$$

Functions on $[-1, 1]$ are extended by zero. Since (1) is the compression of whole-line Gaussian convolution,

$$\langle f, S_u f \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-u\xi^2} |\widehat{f}(\xi)|^2 d\xi. \quad (6)$$

Consider the normalized Dirichlet ground state

$$f_1(p) = \cos(\kappa p), \quad -1 \leq p \leq 1.$$

Its zero extension belongs to $H^1(\mathbb{R})$, has $\|f_1\|_2 = 1$, and direct integration gives

$$\widehat{f}_1(\xi) = \frac{2\kappa \cos \xi}{\kappa^2 - \xi^2}, \quad (7)$$

where the apparent singularities at $\xi = \pm\kappa$ are removable. Also, by Plancherel,

$$\frac{1}{2\pi} \int_{\mathbb{R}} \xi^2 |\widehat{f}_1(\xi)|^2 d\xi = \|f_1'\|_2^2 = A. \quad (8)$$

Lemma 2. *The Rayleigh quotient $q(u) := \langle f_1, S_u f_1 \rangle$ satisfies*

$$q(u) = 1 - Au + \frac{4A}{3\sqrt{\pi}} u^{3/2} + E_q(u), \quad |E_q(u)| \leq Mu^2, \quad (9)$$

for all $u > 0$, with an absolute constant M independent of u .

Proof. Let $h(s) = e^{-s} - 1 + s$. Equations (6)–(8) give

$$q(u) - 1 + Au = \int_{\mathbb{R}} h(u\xi^2) w(\xi) d\xi, \quad w(\xi) = \frac{2\kappa^2}{\pi} \frac{\cos^2 \xi}{(\xi^2 - \kappa^2)^2}. \quad (10)$$

Write $B = 2\kappa^2/\pi$ and subtract the leading algebraic tail:

$$r(\xi) = B \cos^2 \xi \left(\frac{1}{(\xi^2 - \kappa^2)^2} - \frac{1}{\xi^4} \right).$$

The function $\xi^4 r(\xi)$ has removable singularities at $0, \pm\kappa$, is bounded on compact sets, and is $O(\xi^{-2})$ at infinity. Hence

$$J := \int_{\mathbb{R}} |\xi^4 r(\xi)| d\xi < \infty. \quad (11)$$

The elementary bound $0 \leq h(s) \leq s^2/2$ for $s \geq 0$ therefore yields

$$\left| \int_{\mathbb{R}} h(u\xi^2) r(\xi) d\xi \right| \leq \frac{J}{2} u^2. \quad (12)$$

It remains to evaluate the subtracted tail. Scaling $t = \sqrt{u} \xi$ gives

$$\int_{\mathbb{R}} \frac{h(u\xi^2)}{\xi^4} d\xi = u^{3/2} \int_{\mathbb{R}} \frac{e^{-t^2} - 1 + t^2}{t^4} dt = \frac{4\sqrt{\pi}}{3} u^{3/2}. \quad (13)$$

For the last identity, two integrations by parts on $(0, \infty)$ reduce the half-line integral first to $\frac{2}{3} \int_0^\infty (1 - e^{-t^2}) t^{-2} dt$ and then to $\frac{2}{3} \sqrt{\pi}$; doubling gives (13).

The oscillatory half of $\cos^2 \xi = (1 + \cos 2\xi)/2$ is uniformly smaller. Define

$$g(t) = \begin{cases} (e^{-t^2} - 1 + t^2)/t^4, & t \neq 0, \\ 1/2, & t = 0. \end{cases}$$

Then $g'(t) = O(t)$ at zero and $g'(t) = O(t^{-3})$ at infinity, so $g' \in L^1(\mathbb{R})$. One integration by parts after scaling gives

$$\left| \int_{\mathbb{R}} \frac{h(u\xi^2)}{\xi^4} \cos(2\xi) d\xi \right| = u^{3/2} \left| \int_{\mathbb{R}} g(t) \cos(2t/\sqrt{u}) dt \right| \quad (14)$$

$$\leq \frac{1}{2} \|g'\|_1 u^2. \quad (15)$$

Combining (10)–(14), and using $B/2 = A/\pi$, proves (9). $\square$

Since $q(u) = 1 + O(u)$, the uniform elementary estimate $\log(1+z) = z + O(z^2)$ near zero turns (9) into

$$\log q(u) = -Au + \frac{4A}{3\sqrt{\pi}} u^{3/2} + O(u^2). \quad (16)$$

All constants in this statement are uniform for sufficiently small u . The variational principle gives $\lambda(u) \geq q(u)$, and both quantities are positive. After decreasing a fixed $u_1 > 0$ if necessary, (16) therefore implies the signed lower bound

$$R(u) \geq \frac{2A}{3\sqrt{\pi}} u^{3/2} \quad (0 < u < u_1). \quad (17)$$

3 A matching resolvent upper bound

For $x \geq 0$, $e^{-x} \leq (1+x)^{-1}$. Comparing the Fourier multipliers in quadratic forms gives

$$S_u \leq T_u, \quad (18)$$

where T_u is the integral operator on $[-1, 1]$ with kernel

$$T_u(p, q) = \frac{1}{2\sqrt{u}} e^{-|p-q|/\sqrt{u}}. \quad (19)$$

Indeed, (19) is the whole-line convolution kernel of $(1 + u\xi^2)^{-1}$. Thus $\lambda(u) \leq \rho(u)$, where $\rho(u)$ denotes the largest eigenvalue of T_u .

Lemma 3. *Let $\varepsilon = \sqrt{u}$. There is a unique $k \in (0, \kappa)$ such that*

$$k \tan k = \frac{1}{\varepsilon}, \quad (20)$$

and

$$\rho(u) = \frac{1}{1 + uk^2}. \quad (21)$$

Proof. If $T_u f = \rho f$, applying $1 - \varepsilon^2 d^2/dp^2$ in the open interval and differentiating (19) at the endpoints gives

$$-f'' = k^2 f, \quad f'(-1) = \varepsilon^{-1} f(-1), \quad f'(1) = -\varepsilon^{-1} f(1), \quad k^2 = \frac{1 - \rho}{\varepsilon^2 \rho}. \quad (22)$$

The top eigenfunction is unique, strictly positive, and even because the kernel is strictly positive and reflection invariant. Moreover $0 < \rho < 1$: the upper inequality follows strictly from the multiplier comparison with the identity for every nonzero eigenfunction. Hence $k > 0$, and positivity forces $k < \kappa$. The even solution of (22) is a multiple of $\cos(kp)$, so its boundary condition is exactly (20), while the definition of k gives (21). Finally, $k \tan k$ increases from zero to infinity on $(0, \kappa)$, proving uniqueness. $\square$

Set $\delta = \kappa - k$. Equation (20) is equivalent to

$$\tan \delta = \varepsilon k,$$

and therefore

$$0 < \delta \leq \varepsilon k \leq \kappa \varepsilon. \quad (23)$$

Using $0 \leq x - \log(1+x) \leq x^2/2$ for $x \geq 0$, together with (21)–(23), yields, for $0 < u \leq 1$,

$$\begin{aligned} R(u) &\leq \log \rho(u) + Au \\ &= u(A - k^2) + \{uk^2 - \log(1 + uk^2)\} \\ &\leq 2Au^{3/2} + \frac{A^2}{2}u^2 \leq \left(2A + \frac{A^2}{2}\right)u^{3/2}. \end{aligned} \quad (24)$$

Together, (17) and (24) prove (3).

4 Endpoint consequence and limits of the result

The lower bound (17) immediately gives (4). It also follows from (3) that $\Phi(u) = -Au + O(u^{3/2})$, so $\Phi'(0+) = -A$ in the endpoint difference-quotient sense.

For $u > 0$, the simple principal eigenvalue varies smoothly with u . Suppose that the finite classical limit (5) were L . Then

$$R'(u) = Lu + o(u).$$

Integrating from a positive lower endpoint and letting it tend to zero (using $R(u) \rightarrow 0$ from (3)) would give $R(u) = (L/2)u^2 + o(u^2)$, contradicting (4). This completes the proof of Theorem 1.

The trial quotient has the explicit coefficient $4A/(3\sqrt{\pi})$ in (9), whereas the resolvent comparison supplies only a matching order. Thus the argument neither proves a limit for $R(u)/u^{3/2}$ nor identifies an optimized leading coefficient. It also makes no assertion about interchanging endpoint differentiation with the finite-support scaling limit or about separately optimized cumulants in the originating quantum-walk model. The literature comparison made here is bounded to the sources cited; no global priority claim is intended.

Computational and AI assistance disclosure

An AI language model assisted with proof development and manuscript drafting. The mathematical argument was checked through explicit Fourier identities, uniform analytic bounds, and the exactly solved Robin problem displayed above. A separate Python quadrature calculation evaluated the cosine trial quotient and the Robin comparison at five decreasing values of u as a numerical sanity check; no numerical value, fit, or computer output is used in the proof. The author reviewed the final argument and takes responsibility for its contents.

References

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